

Wheat Disease Monitoring in Punjab, Pakistan

Overview

Punjab is a major wheat-producing region in Pakistan, and several foliar diseases have been systematically surveyed over the past two decades. The most frequently reported diseases are spot blotch (caused by *Bipolaris sorokiniana*) and wheat blast (caused by *Magnaporthe oryzae* Triticum pathotype). Monitoring efforts have involved field surveys, seed analyses, and disease index calculations to assess prevalence, incidence, and severity across different agro-ecological zones and crop seasons.

Spot Blotch (*Bipolaris sorokiniana*)

Spot blotch is the most prevalent wheat disease in Punjab. In a 2009 survey, the disease index for spot blotch was 21.8 %. The disease affected 100 % of the Bhakkar-2001 wheat variety. Earlier surveys in 2005, 2006, and 2008 examined eight agro-ecological zones, with Zone 6 being particularly affected.

Wheat Blast (*Magnaporthe oryzae* Triticum pathotype)

Wheat blast, caused by *Magnaporthe oryzae* Triticum pathotype, has been reported in Punjab. Although less frequent than spot blotch, it poses a significant threat due to its rapid spread and potential yield loss. Surveillance has focused on early detection and management practices.

Comparison Table

Disease	Pathogen	Prevalence	Typical Disease Index	Affected Varieties
Spot Blotch	<i>Bipolaris sorokiniana</i>	High	21.8% (2009)	Bhakkar-2001
Wheat Blast	<i>Magnaporthe oryzae</i> Triticum pathotype	Low	Not quantified	Various

References

Sources: academia.edu, researchgate.net, semanticscholar.org, sabraojournal.org, pakbs.org, cabidigitallibrary.org.